

Product Requirements Document for LeadOS

1. Product Overview

1.1 Product Name

LeadOS

1.2 One Line Description

LeadOS is a multilingual AI lead conversion platform that contacts leads through chat or voice, handles objections, scores intent, and routes qualified leads to human relationship managers with full context.

1.3 Product Vision

LeadOS should act as the intelligent conversion layer between marketing campaigns and sales teams.

Instead of letting leads sit in a queue until an RM manually calls them, LeadOS immediately engages each lead, speaks in the lead's preferred language, explains the offer, handles common objections, qualifies the lead, and gives the RM a complete handoff brief.

The MVP will use a fictional demo client called **FinPartner Pro** to demonstrate the product without using Rupeezy branding or IP.

1.4 Product Positioning

LeadOS is not just a chatbot.

It is a full lead conversion operating system:

- Lead capture
- AI conversation
- Objection handling
- Scoring
- Routing
- RM handoff
- Follow up
- Analytics

1.5 MVP Goal

The MVP must prove this flow clearly:

- A lead submits a form
 - AI starts a conversation
 - AI speaks in the selected language
 - AI handles at least one core objection
 - Transcript is saved
 - Scoring classifies the lead as Hot, Warm, or Cold
 - Hot lead appears in RM portal with full context
 - Warm lead gets WhatsApp follow up link
 - Admin dashboard shows funnel and analytics
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2. Problem Statement

2.1 Current Problem

In traditional RM driven lead conversion, the business loses potential partners because the process is slow, inconsistent, and difficult to scale.

Main problems:

2.1.1 Delayed Contact

Leads generated after hours, on weekends, or during campaign spikes are not contacted quickly. By the time an RM reaches out, many leads have gone cold.

2.1.2 RM Capacity Limit

Each RM can only handle one call or conversation at a time. During a campaign surge, the queue grows faster than the team can process it.

2.1.3 Language Mismatch

Many leads are more comfortable in Hindi, Hinglish, or regional language style, but they may receive a formal English call. This reduces trust and engagement.

2.1.4 Inconsistent Objection Handling

Different RMs may handle the same objection differently. Some may answer confidently, while others may miss the right framing.

2.1.5 Poor Handoff Context

When a lead is handed to an RM, the RM often receives basic information like name and phone number, but not the conversation context, objections, interest signals, or recommended next action.

2.1.6 Limited Funnel Visibility

Admins may know how many leads were generated, but not enough about language distribution, objection patterns, AI conversation quality, Hot/Warm/Cold split, or RM follow up performance.

3. Product Objectives

3.1 Primary Objectives

LeadOS MVP must:

1. Capture leads from a dummy client website
2. Engage leads using AI chat and optionally voice
3. Support English, Hindi, and Hinglish
4. Handle the five core sales objections
5. Score every completed conversation
6. Classify leads as Hot, Warm, or Cold
7. Route Hot leads to RMs with full context
8. Generate WhatsApp follow up links for Warm leads
9. Close Cold leads
10. Show admin analytics and RM handoff details

3.2 Business Objectives

LeadOS should help businesses:

1. Reduce time to first contact
2. Improve lead qualification consistency

3. Reduce RM workload on low intent leads
4. Improve Hot lead follow up quality
5. Increase visibility into campaign quality
6. Build a reusable enterprise SaaS platform beyond one client

3.3 Hackathon Objectives

The product should demonstrate:

1. Conversation quality
 2. Multilingual handling
 3. Natural objection handling
 4. Explainable qualification logic
 5. RM handoff design
 6. Dashboard visibility
 7. Practical MVP implementation
-

4. Target Users

4.1 Lead

The external person interested in the partner program.

Needs

1. Wants quick explanation
2. Wants to speak in preferred language
3. Wants answers to objections
4. Wants easy next step
5. Does not want a robotic experience

Product Value

Lead gets an immediate, natural conversation without waiting for an RM.

4.2 Admin

The client side manager.

Needs

1. Wants to see lead funnel
2. Wants Hot/Warm/Cold breakdown
3. Wants to monitor RM performance
4. Wants to understand common objections
5. Wants to manage users and RMs
6. Wants visibility into follow ups

Product Value

Admin gets a central dashboard for AI driven lead conversion operations.

4.3 RM

The relationship manager.

Needs

1. Wants to know which leads are worth calling
2. Wants context before calling
3. Wants to know what the AI already discussed
4. Wants recommended next action
5. Wants a suggested opening line
6. Wants to update outcome easily

Product Value

RM spends time on high intent leads with full context instead of cold calling blindly.

4.4 Product Owner or SaaS Operator

The LeadOS team.

Needs

1. Wants reusable product for multiple clients

2. Wants clean architecture
3. Wants configurable client data
4. Wants scalable AI and voice provider choices
5. Wants demo friendly flows

Product Value

LeadOS can become a reusable enterprise SaaS platform, not a one off hackathon tool.

5. Personas

5.1 Lead Persona: Amit Sharma

Name: Amit Sharma
Role: Financial Advisor
City: Hyderabad
Language: Hinglish
Network: 25 to 30 finance contacts
Concern: Already works with another broker
Goal: Wants to compare benefits and understand onboarding

Amit is likely Hot if the AI handles the existing broker objection well and he agrees to RM callback.

5.2 Lead Persona: Priya Mehta

Name: Priya Mehta
Role: Referrer
City: Mumbai
Language: Hindi
Network: Small but relevant
Concern: Needs time to think
Goal: Wants details on WhatsApp

Priya is likely Warm if she wants details but is not ready for RM callback.

5.3 Lead Persona: Rahul Verma

Name: Rahul Verma
Role: Other
City: Delhi
Language: English
Network: Unclear
Concern: Not interested
Goal: Does not want follow up

Rahul is likely Cold.

5.4 Admin Persona: Sharad

Role: Admin
Goal: Monitor lead conversion pipeline, users, RMs, analytics, follow ups

5.5 RM Persona: Shivansh

Role: Relationship Manager
Goal: Act on assigned Hot leads using AI summary, score reason, transcript

6. Product Scope

6.1 MVP Scope

The MVP includes:

1. LeadOS SaaS Website
2. Dummy Client Website
3. Lead form
4. AI chat conversation
5. Voice request or Vapi voice demo path
6. Language selection
7. Language switching
8. Objection handling
9. Transcript storage

10. Scoring engine
11. Hot/Warm/Cold classification
12. RM task creation
13. WhatsApp wa.me link generation
14. Admin Portal
15. RM Portal
16. Users management
17. Analytics
18. Follow ups
19. Knowledge documents view
20. Feedback collection

6.2 Non MVP Scope

Do not build these unless core MVP is done:

1. Real WhatsApp Business API approval
 2. Billing and subscription
 3. Advanced CRM pipeline
 4. Multi client onboarding UI
 5. Full campaign builder
 6. Prompt approval workflow UI
 7. Model training pipeline
 8. Pipecat production implementation
 9. Mobile app
 10. Advanced data chatbot
 11. Real time socket notifications
 12. Complex audit logs
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7. Product Surfaces

7.1 LeadOS SaaS Website

Public marketing website for LeadOS.

Purpose

Explain the product and route users to the demo.

Main Sections

1. Hero
2. Problem
3. Solution
4. How It Works
5. Features
6. Product Preview
7. Architecture Preview
8. Demo CTA
9. Footer

Key CTA Buttons

1. Try Demo
2. Open Admin Portal
3. Open RM Portal

7.2 Dummy Client Website

Demo website for fictional client FinPartner Pro.

Purpose

Capture leads and start AI conversation.

Main Sections

1. Hero
2. Benefits
3. Lead form
4. Chat window
5. Voice request state
6. Thank you state

7.3 Admin Portal

Client manager dashboard.

Main Pages

1. Dashboard
2. Leads
3. Lead Detail
4. Conversations
5. Analytics
6. Campaigns
7. Users
8. Follow Ups
9. Knowledge Base
10. Feedback
11. Settings

7.4 RM Portal

Relationship manager workspace.

Main Pages

1. Dashboard
2. Hot Leads
3. My Leads
4. Lead Detail
5. Follow Ups
6. Notifications
7. Feedback

8. User Journeys

8.1 Lead Journey

Lead opens dummy client website
→ Reads partner program benefits
→ Fills form
→ Selects language
→ Selects chat or voice

- Gives WhatsApp consent
- Starts conversation
- AI explains opportunity
- Lead asks question or raises objection
- AI responds naturally
- Lead gives next step
- Conversation ends
- Lead sees thank you message

8.2 Hot Lead Journey

- Lead completes conversation
- Scoring marks lead Hot
- System creates RM task
- RM receives notification
- RM opens lead detail
- RM reads transcript and score reason
- RM uses suggested opening line
- RM calls lead
- RM marks converted or closed

8.3 Warm Lead Journey

- Lead completes conversation
- Scoring marks lead Warm
- System creates WhatsApp follow up
- Admin/RM sees message preview
- RM opens WhatsApp link
- RM marks follow up sent

8.4 Cold Lead Journey

- Lead completes conversation
- Scoring marks lead Cold
- System closes lead
- Admin can still review reason and transcript
- No RM task is created

8.5 Admin Journey

Admin logs in

- Views dashboard
- Checks lead funnel
- Opens leads list
- Filters Hot leads
- Opens lead detail
- Reviews transcript and score
- Checks users and RMs
- Checks follow ups
- Reviews analytics and feedback

8.6 RM Journey

RM logs in

- Views Hot lead queue
 - Opens lead detail
 - Reads AI handoff brief
 - Understands objection and next action
 - Calls lead
 - Marks task converted or closed
 - Submits feedback if score was wrong
-

9. Feature Requirements

9.1 Lead Capture

Description

The system must capture leads from the dummy client website.

Requirements

1. Lead form must validate required fields
2. Lead must be attached to client

3. Lead must be attached to campaign
4. Preferred language must be stored
5. Preferred channel must be stored
6. WhatsApp consent must be stored
7. Public lead session token must be created

Acceptance Criteria

1. Valid form creates a lead
 2. Invalid form shows validation errors
 3. Lead appears in Admin Portal
 4. Lead can start chat after form submission
-

9.2 AI Chat Conversation

Description

The system must allow leads to chat with AI.

Requirements

1. Chat starts after lead creation
2. AI greeting uses selected language
3. User messages are saved
4. Assistant messages are saved
5. AI uses approved conversation structure
6. AI handles objections naturally
7. Chat can be ended
8. Transcript is compiled

Acceptance Criteria

1. Lead can send message
 2. AI responds
 3. Transcript appears in Admin and RM portals
 4. Conversation status becomes completed after ending
-

9.3 AI Voice Conversation

Description

The system should support a voice path through Vapi if ready.

Requirements

1. Lead can request voice
2. System creates voice conversation
3. Vapi transcript events can be received
4. Transcript is saved
5. Call ended event triggers scoring

Acceptance Criteria

1. Voice UI does not break
 2. Voice fallback to chat exists
 3. Vapi webhook path can be demonstrated or explained
-

9.4 Multilingual Handling

Description

The AI must support English, Hindi, and Hinglish.

Requirements

1. Use selected language first
2. Detect language switch
3. Save final language preference
4. Ask language if uncertain
5. Fall back to Hindi or English if unsupported

Acceptance Criteria

1. Hinglish selected lead gets Hinglish greeting
 2. User can switch language
 3. Language appears in lead detail and analytics
-

9.5 Objection Handling

Description

The AI must handle the five required objections.

Required Objections

1. Existing broker
2. Not enough contacts
3. Client support concern
4. Trust concern
5. Think later or call later

Requirements

1. AI must acknowledge concern
2. AI must answer using approved context
3. AI must personalize response
4. AI must ask next step
5. Scoring must extract objection and resolution status

Acceptance Criteria

1. Existing broker objection is handled naturally
 2. Objection appears in lead detail
 3. Objection resolution status appears
 4. Objection analytics count updates
-

9.6 Scoring Engine

Description

After conversation completion, the system scores the lead.

Requirements

1. Score readiness out of 40

2. Score engagement out of 35
3. Score fit out of 25
4. Total score out of 100
5. Classify as Hot, Warm, or Cold
6. Generate reason
7. Generate recommended next action
8. Generate suggested RM opening line for Hot leads
9. Extract objections
10. Save scoring output

Acceptance Criteria

1. Hot lead gets score 70 or more or valid override
 2. Warm lead gets follow up
 3. Cold lead closes
 4. Score reason is visible
 5. Score breakdown is visible
-

9.7 Routing

Description

System routes leads based on classification.

Hot

Requirements:

1. Assign RM
2. Create RM task
3. Create notification
4. Update status to assigned_to_rm

Warm

Requirements:

1. Generate WhatsApp message
2. Generate wa.me link
3. Create follow up

4. Update status to follow_up_scheduled

Cold

Requirements:

1. Update status to closed
2. Do not create RM task
3. Do not create follow up by default

Acceptance Criteria

1. Hot appears in RM portal
 2. Warm appears in Follow Ups page
 3. Cold appears as closed
-

9.8 Admin Dashboard

Description

Admin dashboard shows complete lead conversion overview.

Requirements

KPI cards:

1. Total Leads
2. Conversations Completed
3. Hot Leads
4. Warm Leads
5. Cold Leads
6. Assigned to RM
7. Follow Ups Ready
8. Converted

Charts:

1. Funnel
2. Classification split
3. Language distribution
4. Objection distribution

Acceptance Criteria

1. Dashboard loads
 2. Metrics match backend data
 3. Hot/Warm/Cold distribution is visible
 4. Language and objection insights are visible
-

9.9 Admin Lead Detail

Description

Admin can inspect complete lead intelligence.

Requirements

Show:

1. Profile
2. Language
3. Channel
4. WhatsApp consent
5. Status
6. Classification
7. Score breakdown
8. AI summary
9. Objections
10. Transcript
11. RM assignment
12. Follow up
13. Suggested opening line

Acceptance Criteria

1. Admin can understand why the lead got its score
 2. Transcript is readable
 3. Objections are clear
 4. Follow up or RM task is visible
-

9.10 Users Management

Description

Admin can manage Admin and RM users.

Requirements

1. List users
2. Filter by role
3. Filter by active status
4. Add RM
5. Deactivate RM
6. View assigned lead count
7. View pending task count
8. View converted count

Acceptance Criteria

1. Admin can add RM
 2. Admin can deactivate RM
 3. RM appears in assignment data
-

9.11 RM Handoff

Description

RM must receive full context for Hot leads.

Requirements

RM lead detail must show:

1. Lead profile
2. Score
3. Score reason
4. AI summary
5. Objections
6. Transcript

7. Recommended next action
8. Suggested RM opening line
9. Task status

Acceptance Criteria

1. RM can continue without asking lead to repeat everything
 2. RM can mark converted or closed
 3. RM can submit feedback
-

9.12 Follow Ups

Description

Warm leads should produce WhatsApp follow up links.

Requirements

1. Generate message preview
2. Generate wa.me link
3. Show status ready/opened/sent/cancelled
4. Allow Open WhatsApp
5. Allow Mark Sent

Acceptance Criteria

1. Warm lead creates follow up
 2. WhatsApp link opens
 3. Status updates to sent
-

9.13 Feedback

Description

RM can submit feedback about scoring.

Requirements

1. Select corrected classification
2. Enter corrected score optional
3. Select feedback type
4. Add comment
5. Save feedback

Acceptance Criteria

1. Feedback appears in admin feedback page
 2. Feedback does not auto deploy scoring changes
-

10. Data Model Summary

10.1 Main Entities

1. Client
2. User
3. Campaign
4. Lead
5. Conversation
6. Conversation Message
7. Lead Score
8. RM Task
9. Follow Up
10. Notification
11. Knowledge Document
12. RM Feedback

10.2 Lead Lifecycle

```
new
→ pending_contact
→ conversation_completed
→ assigned_to_rm / follow_up_scheduled / closed
→ converted / closed
```

10.3 Classification Lifecycle

No classification

→ hot / warm / cold

10.4 Routing Mapping

Hot → RM Task

Warm → WhatsApp Follow Up

Cold → Closed

11. API Product Requirements

11.1 API Principle

All frontends must call APIs only.

Frontend must not call Supabase DB directly.

11.2 Required API Groups

1. Auth APIs
2. Public Lead APIs
3. Lead APIs
4. Chat APIs
5. Voice APIs
6. Scoring APIs
7. RM Task APIs
8. Follow Up APIs
9. Notification APIs
10. Analytics APIs
11. User APIs
12. Knowledge APIs
13. Feedback APIs

11.3 API Response Rule

All APIs must return standard response format:

```
{  
  "success": true,  
  "data": {}  
}
```

or:

```
{  
  "success": false,  
  "error": {  
    "code": "ERROR_CODE",  
    "message": "Readable error message"  
  }  
}
```

12. Analytics Requirements

12.1 Admin Analytics

Admin must see:

1. Total leads
2. Conversations completed
3. Hot leads
4. Warm leads
5. Cold leads
6. Assigned to RM
7. Follow ups scheduled
8. Converted leads
9. Language distribution
10. Objection distribution
11. RM performance
12. Campaign performance

12.2 RM Analytics

RM must see:

1. Assigned Hot leads
2. Pending tasks
3. Follow ups due
4. Converted leads
5. Closed leads

12.3 Objection Analytics

System must count:

1. Existing broker
2. Not enough contacts
3. Client support concern
4. Trust concern
5. Call later
6. Other

For each objection, show:

1. Count
 2. Resolved
 3. Partially resolved
 4. Unresolved
-

13. Knowledge Requirements

13.1 Knowledge Documents

MVP must support seeded knowledge documents:

1. Sales script
2. FAQ
3. Onboarding guide
4. Objection guide

13.2 Knowledge Usage

Knowledge documents are used by AI to answer factual questions and avoid hallucination.

13.3 Knowledge Editing

Editing is optional for MVP.

Viewing seeded documents is enough.

14. Notifications Requirements

14.1 Notification Types

The system must support:

1. New Hot Lead Assigned
2. Follow Up Ready
3. Follow Up Due
4. Score Ready
5. Task Updated

14.2 Notification Behavior

Users must be able to:

1. View notifications
 2. Mark one notification as read
 3. Mark all notifications as read
 4. Open related lead from notification
-

15. Security and Privacy Requirements

15.1 Access Control

1. Admin can access all data for client
2. RM can access only assigned lead data
3. Public lead can access only own chat session
4. APIs must validate role and client scope

15.2 Data Privacy

The system must avoid collecting sensitive financial credentials.

AI must not ask for:

1. OTP
2. Password
3. Bank credentials
4. Sensitive investment account details

15.3 Secret Management

Frontend must never receive:

1. Supabase service role key
2. OpenAI API key
3. Vapi API key
4. Webhook secret

15.4 RLS

RLS must be enabled on Supabase tables.

16. Non Functional Requirements

16.1 Performance

1. Lead list must be paginated
2. Dashboard should load quickly
3. Chat should show loading state
4. Large transcripts should not freeze UI
5. Analytics should not block main pages

16.2 Reliability

1. Lead must not be lost if AI fails
2. Transcript must be saved before scoring
3. Scoring fallback must exist
4. Voice failure must fall back to chat
5. WhatsApp link failure must not break lead detail

16.3 Usability

1. UI must be clean and readable
2. Badges must make statuses obvious
3. RM handoff must be easy to understand
4. Admin dashboard must be demo friendly
5. Lead form must be simple

16.4 Maintainability

1. Use TypeScript types
2. Use reusable components
3. Keep API services separate
4. Keep AI prompts separate
5. Keep scoring schema separate
6. Use clear constants for statuses and roles

16.5 Scalability

The system should later support:

1. More clients
 2. More languages
 3. More RMs
 4. More campaigns
 5. Real WhatsApp API
 6. Pipecat scale path
 7. Better RAG and memory
 8. Provider switching
-

17. MVP Success Metrics

For hackathon/demo, success is not measured by real production conversion yet.

MVP success metrics:

1. Time from lead submission to AI conversation start
2. Percentage of conversations completed
3. Hot/Warm/Cold classification generated
4. RM handoff created for Hot lead
5. WhatsApp follow up created for Warm lead
6. Transcript visible in portals
7. Score reason visible in portals
8. Objection extracted correctly
9. Full demo flow completes without manual DB edits

Demo target:

```
One Hot flow working
One Warm flow working
One Cold flow working
Admin portal working
RM portal working
```

18. MVP Prioritization

18.1 Priority 1

Must build:

1. Dummy client lead form
2. Public lead creation
3. Chat conversation
4. Transcript storage
5. Scoring
6. Routing
7. Admin lead list
8. Admin lead detail

9. RM Hot lead detail
10. Follow up link generation

18.2 Priority 2

Should build:

1. SaaS website
2. Admin dashboard
3. Users page
4. Analytics page
5. Follow ups page
6. Notifications
7. Feedback
8. Knowledge view

18.3 Priority 3

Nice to have:

1. Voice Vapi live demo
 2. Campaign creation
 3. Data chatbot
 4. Advanced charts
 5. More animations
 6. n8n workflow demo
 7. Prompt tuning UI
-

19. Release Plan

19.1 MVP Build Phase

Build:

1. Backend database and APIs
2. Dummy client website
3. Admin portal
4. RM portal

5. AI chat
6. Scoring
7. Routing

19.2 Demo Stabilization Phase

Focus on:

1. Fixing main demo bugs
2. Adding seeded data
3. Polishing Hot/Warm/Cold flows
4. Testing role access
5. Preparing fallback flows

19.3 Presentation Phase

Prepare:

1. Primary chat demo
 2. Warm follow up demo
 3. RM handoff explanation
 4. Admin analytics walkthrough
 5. Voice architecture explanation or live demo if stable
-

20. Dependencies

20.1 Internal Dependencies

1. Backend must define API contracts
2. Admin portal must match API data shape
3. RM portal must match API data shape
4. Dummy client website must create leads correctly
5. Scoring must run before routing
6. Routing must create RM task or follow up

20.2 External Dependencies

1. Supabase project
 2. OpenAI API key
 3. Vapi account if voice demo is used
 4. WhatsApp wa.me support
 5. Deployment platform
-

21. Risks and Mitigations

21.1 Risk: Voice Integration Takes Too Long

Mitigation:

Use chat as primary live demo and show voice UI plus webhook architecture.

21.2 Risk: AI Hallucinates

Mitigation:

Use strict prompt, approved knowledge, and fallback to RM confirmation.

21.3 Risk: Scoring Fails

Mitigation:

Use fallback Warm score and manual review reason.

21.4 Risk: Frontend and Backend Data Shapes Mismatch

Mitigation:

Use shared API contracts and mock services that match backend response.

21.5 Risk: WhatsApp API Not Approved

Mitigation:

Use wa.me link generation for MVP.

21.6 Risk: Too Many Features

Mitigation:

Freeze scope around the core flow:

Lead → AI conversation → transcript → scoring → routing → RM handoff/foll

22. Open Decisions

These can be finalized during implementation:

1. Whether Vapi live voice will be part of live demo or architecture only
 2. Whether n8n will be actively used or explained as workflow layer
 3. Whether knowledge documents are editable or only seeded
 4. Whether Admin and RM portals will be separate deployments or route groups in same app
 5. Whether user creation through Supabase Auth is built or seeded manually for MVP
-

23. Final Acceptance Checklist

The product is ready when:

1. LeadOS website opens
2. Dummy client website opens
3. Lead form creates lead
4. Chat starts
5. AI handles objection
6. Transcript is stored
7. Scoring runs
8. Hot lead creates RM task
9. Warm lead creates WhatsApp follow up
10. Cold lead closes
11. Admin dashboard shows metrics
12. Admin lead detail shows full AI context
13. RM portal shows assigned Hot lead
14. RM lead detail shows full handoff brief
15. RM can mark lead converted or closed

- 16. Follow up link opens WhatsApp
 - 17. Users page shows Admins and RMs
 - 18. Role access works
 - 19. Demo works end to end
 - 20. Seeded fallback data exists
-

24. Final Product Definition

LeadOS MVP is complete when it can confidently show this:

A business can upload or capture leads, let AI engage them immediately ir

That is the core product.